YAOYUN ZHANG

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Research Interests

My research interests mainly lie in **universal audio intelligence**, with an emphasis on multimodal audio generation and neural audio representation learning. I am particularly interested in developing unified audio codecs and generative models that support speech, music, and general sounds while **remaining efficient, robust, and generalizable across modalities and domains**.

Education

School of Computer Science, Shanghai Jiao Tong Univerisity *Shanghai, China*
· M.Eng. in Computer Technology, Advised by Prof. Mengyue Wu Sep. 2023 - Mar. 2026
Grades (Cumulative GPA): 3.11/4.0

Chengdu University of Technology *Chengdu,China*
◇ Dual Bachelor's Degrees Sep. 2019 - Jun. 2023
· B.Eng. in Computer Science and Technology, School of Computer Science
· B.A. in English, College of Foreign Languages and Cultures
Grades (Cumulative GPA): 3.01/4.0

Research Experience

Meituan, Beijing Sankuai Online Technology Co., Ltd. *Beijing, China*
Research Intern — Mentors: Guoqiao Yu, Fengyu Yang, and Shansong Liu Jun 2025 - Sep 2025

◇ **Audio Generation and Neural Codec**

- Developed BEST-RQ-based neural audio codecs and tokenizers for unified representation of speech, music, and general sounds, supporting audio drama and music generation.
- Designed a three-stage training framework combining speech pretraining, multimodal adaptation, K-means-initialized vector quantization, and Flow-Matching DiT reconstruction; processed nearly one million hours of audio data.
- Improved low-bitrate reconstruction through multi-codebook quantization, diversity regularization, and local feature fusion, achieving consistent gains in speech intelligibility and audio reconstruction quality.

X-LANCE Lab, Shanghai Jiao Tong University *Shanghai, China*
Graduate Researcher — Advisor: Prof. Mengyue Wu Sep 2023 - Mar 2026

◇ **Conducted research on multimodal audio generation, universal audio representation learning, and trustworthy audio intelligence.**

- Led Smooth-Foley (ICASSP 2025), introducing frame-level visual fusion and semantic temporal conditioning for continuous and synchronized video-to-audio generation.
- Contributed to UniFlow-Audio, a unified flow-matching framework for both time-aligned and non-time-aligned audio generation tasks, including video-to-audio and text-to-audio tasks.
- Co-developed FakeSound2 (INTERSPEECH 2026), an explainable and generalizable deepfake sound detection benchmark covering manipulation localization, forgery attribution, and out-of-domain evaluation.

Publications

Smooth-Foley: Creating Continuous Sound for Video-to-Audio Generation Under Semantic Guidance
Yaoyun Zhang, Xuenan Xu, Mengyue Wu*.
ICASSP 2025 | Smooth-Foley.github.io | ArXiv:2412.18157

FakeSound2: A Benchmark for Explainable and Generalizable Deepfake Sound Detection
Zeyu Xie, Yaoyun Zhang, Xuenan Xu, Yongkang Yin, Chenxing Li, Mengyue Wu*, Yuexian Zou*
INTERSPEECH 2026 | FakeSound2.github.io | ArXiv:2509.17162

UniFlow-Audio: A Unified Flow Matching Framework for Audio Generation from Omni-Modalities
Xuenan Xu, Jiahao Mei, Zihao Zheng, Ye Tao, Zeyu Xie, Yaoyun Zhang, Haohe Liu, Yuning Wu, Ming Yan, Wen Wu, Chao Zhang, Mengyue Wu*.
ArXiv 2025 | UniFlow-Audio.github.io | ArXiv:2509.24391

Domain Invariant Learning for Gaussian Processes and Bayesian Exploration
Zhaoxi Long, Siyuan Bian, Yaoyun Zhang, Yuliang Zhang, Qinying Gu, Xinbing Wang, Chenghu Zhou, Nanyang Ye*.
AAAI 2024 | DIL-GP.github | ArXiv:2312.11318

Honor & Awards

Scholarship
National Encouragement Scholarship (Top 3% nationwide) *2020*

Competition
Second Prize, National Computer Game Competition *2021*

Skills

Skills: Python (Pytorch), html, css, Linux, LaTeX
Languages: English (CATTI International Certificate C-Level), Chinese (Native)

References

- Mengyue Wu, Associate Professor of Computer Science, Shanghai Jiao Tong University
- Guoqiao Yu, Principal Researcher, Meituan Co., Speech Synthesis Department
- Fengyu Yang, Senior Researcher, Meituan Co., Speech Synthesis Department
- Shansong Liu, Senior Researcher, TeleAI, China Telecom
- Nanyang Ye, Associate Professor of Computer Science, Shanghai Jiao Tong University